Module 04 Lab Exercise - Contribution Journal

ITAI 1371 - Introduction to Machine Learning

John Grubbs

I independently created and ran my copy of the Module 4 Exploratory Data Analysis lab and completed the assignment work in a Python (.py) file. I worked through the provided code to understand how Pandas, Matplotlib, and Seaborn are used to explore and visualize the Titanic dataset.

I examined the dataset's structure, missing values, and summary statistics, then reviewed the plots to identify patterns related to passenger class, sex, age, and survival. I also wrote additional visualization code to explore embarkation port and passenger fare, developing my own observations from the results.

I created and managed the Git repository for the lab to organize my files and track changes. This supported my independent coding and analysis work throughout the assignment.

Keira Burdine

My contribution to this lab included completing and troubleshooting the Google Colab notebook, running the required exploratory data analysis, reviewing the outputs, completing the student experimentation section, and answering the knowledge-check questions.

I began by loading and examining the Titanic dataset using Pandas. I ran the commands that displayed the first five rows of the dataset and reviewed the dataset information. I identified the number of rows and columns, the different data types, and columns containing missing values. I then generated descriptive statistics and reviewed values such as the average passenger age, survival rate, and range of passenger fares. I also completed the visual exploratory analysis. I generated and reviewed visualizations showing the overall survival distribution, survival based on passenger class, survival based on gender, and the relationship between age and survival. I examined each graph to understand what the visualization demonstrated rather than only running the code. From these plots, I observed that passenger class and gender showed strong relationships with survival.

During the lab, I encountered several syntax errors because multiple sections of the provided notebook had Python commands combined onto a single line. I corrected these sections by separating the commands into the proper Python format. I also corrected explanatory text that had been interpreted as Python code by converting it into comments. After making these corrections, I reran the cells and verified that the outputs and visualizations displayed successfully.

For the student experimentation section, I wrote code to create a countplot comparing survival by port of embarkation. This allowed me to compare passengers who boarded at Southampton,

Cherbourg, and Queenstown. I then created a boxplot to compare passenger fares based on whether the passenger survived. I reviewed the resulting plots and used them to make observations about relationships within the dataset. Finally, I completed the knowledge-check portion of the notebook by explaining the purpose of EDA, identifying the type of passenger who appeared to have the strongest chance of survival, and explaining why visualizations provide information that summary statistics alone may not show.

My overall contribution consisted of completing the required notebook work, correcting coding errors, creating the required experimental visualizations, interpreting the results, and verifying that the notebook executed correctly.

Luis Adrian

My contribution to this lab was running the provided code cells and reviewing their outputs. I worked through the summary statistics and the visualizations about survival, passenger class, gender, and age. And making sure that the code is running as expected.

And for the experimentation section I created a countplot to compare survival by port of embarkation and explored fare differences between survivors and non-survivors. And answering the the knowledge section question that ask about what I learn in this lab.